empowers people with CKD and supports self-care management. People put a large value on diet, cultural preferences, and QoL; however, adherence to a low-protein diet remains challenging, may impact social and psychological well-being, and given that most of the trials for protein restriction were conducted before RASi and SGLT2i were implemented, may not be worth the sacrifice/change in lifestyle. The impact of protein restriction and the use of non–animal-based protein diets should be evaluated in the context of new care paradigms to ascertain the incremental gain of these strategies relative to the efforts and costs.

Special considerations

Pediatric considerations.

Practice Point 3.3.1.4: Do not restrict protein intake in children with CKD due to the risk of growth impairment. The target protein and energy intake in children with CKD G2–G5 should be at the upper end of the normal range for healthy children to promote optimal growth.

Children with CKD likely have similar resting energy expenditure to healthy children and should have total energy requirements in the normal range.[483] As in adults, protein restriction was considered for children with CKD in the past. Two RCTs have compared low-protein versus normal-protein diets in children with CKD.[484,485] One found poorer growth for those on a low-protein diet, and the other found no difference in eGFR between the groups. A 2007 Cochrane meta-analysis concluded that there was uncertainty over the possible harm of strict low-protein diets on growth in young infants.[486] The 2009 KDOQI guidelines and the 2020 Pediatric Renal Nutrition Taskforce suggest maintaining an intake of dietary protein at 100%–140% of the dietary reference intake (DRI) or the SDI for ideal body weight in children with CKD G3 and at 100%–120% of the DRI/SDI in children with CKD G4–G5.[487,488]

Older adults.

Practice Point 3.3.1.5: In older adults with underlying conditions such as frailty and sarcopenia, consider higher protein and calorie dietary targets.

In older adults with CKD, nutritional management should consider potential challenges stemming from simultaneous and potentially conflicting risks of CKD progression and malnutrition/protein-energy wasting. In older adults, protein targets should be set after careful individual assessment to identify the most urgent clinical challenge.

Geriatric guidelines recommend protein intakes of 1.0–1.2 g/kg body weight/d to prevent age-related malnutrition and prevent sarcopenia.[489] Such protein intakes may be appropriate in some people with stable or slowly progressing CKD, whose clinical picture is dominated by old age and related challenges to their nutritional and functional status. On the other hand, protein restriction may be appropriate in older adults whose primary clinical challenge is CKD with significant progression, provided they are metabolically stable.[490] The course of action should consider patient preferences and when necessary, involve family members and caregivers.

3.3.2 Sodium intake

The Work Group concurs with the following recommendation from the KDIGO 2022 Clinical Practice Guideline for Diabetes Management in Chronic Kidney Disease[23] and the KDIGO 2021 Clinical Practice Guideline for the Management of Blood Pressure in Chronic Kidney Disease.[21]

Recommendation 3.3.2.1: We suggest that sodium intake be <2 g of sodium per day (or <90 mmol of sodium per day, or <5 g of sodium chloride per day) in people with CKD (*2C*).

Practice Point 3.3.2.1: Dietary sodium restriction is usually not appropriate for patients with sodium-wasting nephropathy.

Global average sodium intake is estimated to be 4310 mg/d (10.78 g of salt per day), which far exceeds the physiological requirement and is more than double the WHO recommendation of <2 g of sodium (equivalent to <5 g of salt) per day in adults.[491] This perhaps reflects the pervasive use of sodium in many commercial food products, which makes achieving WHO targets challenging to meet for many people. There are large-scale RCTs quantifying the benefits of restricted salt intake (e.g., using 75% sodium and 25% potassium chloride salt substitutes) to lower BP and reduce the risk of cardiovascular events in the general population.[492] In RCTs with up to 36 weeks of follow-up, reduction in dietary sodium has also been shown to lower BP and levels of albuminuria in people with CKD.[492–494] Although presumed to reduce the risk of CKD progression and CVD, longer term trials have not been conducted to confirm these effects translate into reduced risk of clinical outcomes in CKD.[493] Given the effects of sodium restriction on BP, it is reasonable to recommend sodium restriction to people with CKD in combination with pharmacological strategies to minimize the risk of kidney and CVDs. People with CKD may have salt-wasting kidney disease, malnutrition, or be exposed to extremely hot climatic conditions. In such scenarios, this recommendation may not apply.

Special considerations

Pediatric considerations.

Practice Point 3.3.2.2: Follow age-based Recommended Daily Intake when counseling about sodium intake for children with CKD who have systolic and/or diastolic blood pressure >90th percentile for age, sex, and height.

The WHO recommends that the maximum intake of <2 g/d sodium (<5 g/d salt) in adults should be adjusted downward based on the energy requirements of children relative to those of adults (Table 22[495]). Children born with low birth weight (<2.5 kg) are at increased risk for CKD in later life and may also be at higher risk for hypertension and increased salt sensitivity. Salt sensitivity is a physiological trait by which BP in some people exhibits changes parallel to changes in salt intake. Children born with low birth weight may have a 37% increased salt sensitivity (defined as an increase in mean BP

Table 22 | Age-based sodium intake recommendations[495]

Age	Recommended adequate sodium intake (g/d)
0–6 mo	0.110
7–12 mo	0.370
1–3 yr	0.370
4–8 yr	1.0
9–13 yr	1.2
14–70 yr	1.5

≥3 mm Hg over 24 hours while on a high salt diet, when compared with a controlled salt diet). That sensitivity may increase further in those who are small for gestational age.[496]

Children with CKD often have underlying tubular conditions that predispose them to numerous electrolyte losses, including sodium. For these children, a supplemented rather than restricted sodium intake will be required. For non–salt-wasting children, salt intake should be limited to the age-based Recommended Daily Intake.

3.4 Blood pressure control

The Work Group concurs with the KDIGO 2021 Clinical Practice Guideline for the Management of Blood Pressure in Chronic Kidney Disease, which encourages individualized BP targets and use of agents according to age, coexistent CVD, and other comorbidities; risk of progression of CKD; and tolerance to treatments.[21] We highlight the following guidance:

Recommendation 3.4.1: We suggest that adults with high BP and CKD be treated with a target systolic blood pressure (SBP) of <120 mm Hg, when tolerated, using standardized office BP measurement *(2B)*.

Practice Point 3.4.1: Consider less intensive BP-lowering therapy in people with frailty, high risk of falls and fractures, very limited life expectancy, or symptomatic postural hypotension.

RCTs have not demonstrated that intensive BP-lowering results in meaningful reductions in the risk of kidney failure, but the RCT evidence supporting important cardiovascular benefits should encourage such a strategy. By aiming for an SBP <120 mm Hg, more adults with CKD will achieve an SBP <130 mm Hg, even if they do not meet the <120 mm Hg target. Observationally, on average, each 20 mm Hg higher than usual SBP and 10 mm Hg higher diastolic BP is associated with an approximate doubling of cardiovascular risk, with no lower limit down to at least 115/75 mm Hg.[497] Data from the Systolic Blood Pressure Intervention Trial (SPRINT) support the SBP target of <120 mm Hg (when measured using a standardized office BP measurement) to reduce cardiovascular risk in adults aged >75 years, or aged >50 years with 1 or more of the following risk factors: clinical or subclinical CVD (other than stroke), eGFR 20–60 ml/min per 1.73 m^2, or ≥15% 10-year cardiovascular risk.[498] Compared with a target of 140 mm Hg, this approach reduces the risk of major adverse cardiovascular events (MACE) by one-quarter (hazard ratio [HR]: 0.75; 95% CI: 0.64–0.89), and that relative benefit was similar in people with and without CKD. The SPRINT trial excluded people with diabetes, but cardiovascular benefits of intensive BP lowering on risk of stroke and heart failure are clearly apparent in people with diabetes in individual patient level data meta-analysis of intensive versus standard BP-lowering trials.[499]

Standardized BP monitoring can be challenging to offer in a clinic setting due to the time required[500]; however, it is considered potentially hazardous to apply the recommended SBP target of <120 mm Hg to BP measurements obtained in a nonstandardized manner.[500] A practical solution to ensure the identification of high BP is by using home-based monitoring (or telemonitoring). Trials have shown that 2 morning and evening BP measurements taken during the first week of every month can be used to titrate antihypertensive medication and reduce BP more than "usual care" approaches.[501]

People who are frail, have limited life expectancy, or have a history of falls and fractures may have increased risk of additional events if BP targets of <120 are achieved. Postural hypotension in these people is associated with adverse outcomes, and thus weighing the benefits of some attenuation of eGFR decline versus the life-changing impact of falls, fractures, and other events should be considered in choosing specific targets.

Special considerations

Pediatric considerations. The Work Group concurs with the KDIGO 2021 Clinical Practice Guideline for the Management of Blood Pressure in Chronic Kidney Disease, and we highlight the following guidance[21]:

Recommendation 3.4.2: We suggest that in children with CKD, 24-hour mean arterial pressure (MAP) by ambulatory blood pressure monitoring (ABPM) should be lowered to ≤50th percentile for age, sex, and height *(2C)*.

Practice Point 3.4.2: Monitor BP once a year with ABPM and every 3–6 months with standardized auscultatory office BP in children with CKD.

Practice Point 3.4.3: In children with CKD, when ABPM is not available, it is reasonable to target manual auscultatory office SBP, obtained in a protocol-driven standardized setting, of 50th–75th percentile for age, sex, and height unless achieving this target is limited by signs or symptoms of hypotension.

These statements with respect to children are generally worded to maintain consistency with the KDIGO 2021 Clinical Practice Guideline for the Management of Blood Pressure in Chronic Kidney Disease,[21] where the full rationale and evidence behind the statements are available. However, the suggestion to target auscultatory office SBP at the 50th–75th percentile when ABPM is not available departs from

the BP guideline (the previous guideline suggested a target <90th percentile). Although office BP may be higher than BP measured by ABPM, this is not universally the case. Given the evidence that intensive BP control may slow CKD progression together with the very low risk of adverse effects of intensive BP lowering in children,[228] we consider that more intensive BP lowering targeting around the 50th percentile is reasonable. However, a target even lower than the 50th percentile has not been shown to offer additional benefits. Recent trial data found that using a target of office auscultatory SBP at 50th to 75th percentile versus intensive control to below the 40th percentile did not result in significant differences in left ventricular mass index.[229]

3.5 Glycemic control

Please refer to the KDIGO 2022 Clinical Practice Guideline for Diabetes Management in Chronic Kidney Disease for specific recommendations, practice points, and research recommendations.

3.6 Renin-angiotensin system inhibitors

The Work Group highlights recommendations from the KDIGO 2021 Clinical Practice Guideline for the Management of Blood Pressure in Chronic Kidney Disease and selected practice points for treatment with RASi from the KDIGO 2021 Clinical Practice Guideline for the Management of Blood Pressure in Chronic Kidney Disease[21] and the KDIGO 2022 Clinical Practice Guideline for Diabetes Management in Chronic Kidney Disease.[23] The Work Group considers several recommendations to apply even in the absence of high BP and has adapted the recommendations from the BP guideline to remove this requirement. Key recommendations and practice points are highlighted:

Recommendation 3.6.1: We recommend starting renin-angiotensin-system inhibitors (RASi) (angiotensin-converting enzyme inhibitor [ACEi] or angiotensin II receptor blocker [ARB]) for people with CKD and severely increased albuminuria (G1–G4, A3) without diabetes *(1B)*.

Recommendation 3.6.2: We suggest starting RASi (ACEi or ARB) for people with CKD and moderately increased albuminuria (G1–G4, A2) without diabetes *(2C)*.

Recommendation 3.6.3: We recommend starting RASi (ACEi or ARB) for people with CKD and moderately-to-severely increased albuminuria (G1–G4, A2 and A3) with diabetes *(1B)*.

Recommendation 3.6.4: We recommend avoiding any combination of ACEi, ARB, and direct renin inhibitor (DRI) therapy in people with CKD, with or without diabetes *(1B)*.

Practice Point 3.6.1: RASi (ACEi or ARB) should be administered using the highest approved dose that is tolerated to achieve the benefits described because the proven benefits were achieved in trials using these doses.

Practice Point 3.6.2: Changes in BP, serum creatinine, and serum potassium should be checked within 2–4 weeks of initiation or increase in the dose of a RASi, depending on the current GFR and serum potassium.

Practice Point 3.6.3: Hyperkalemia associated with use of RASi can often be managed by measures to reduce the serum potassium levels rather than decreasing the dose or stopping RASi.

Practice Point 3.6.4: Continue ACEi or ARB therapy unless serum creatinine rises by more than 30% within 4 weeks following initiation of treatment or an increase in dose.

Practice Point 3.6.5: Consider reducing the dose or discontinuing ACEi or ARB in the setting of either symptomatic hypotension or uncontrolled hyperkalemia despite medical treatment, or to reduce uremic symptoms while treating kidney failure (estimated glomerular filtration rate [eGFR] <15 ml/min per 1.73 m^2).

Practice Point 3.6.6: Consider starting people with CKD with normal to mildly increased albuminuria (A1) on RASi (ACEi or ARB) for specific indications (e.g., to treat hypertension or heart failure with low ejection fraction).

The role of RASi (specifically ACEi or ARB) in the management of BP and people with CKD, diabetes, and/or high BP have been specifically considered in recent KDIGO guidelines.[21,23] Although temporarily stopping RASi may be a valid treatment strategy for emergent hyperkalemia, we advise to ensure the reinitiation of treatments once the adverse event is resolved, so that people are not deprived of a needed medication (Practice Point 4.3.3).[502–506] The Work Group offers the new Practice Point 3.6.6 and a revised algorithm for initiation of RASi (Figure 21).[23] The algorithm has been updated to suggest a ≥30% decrease in eGFR (rather than increase in creatinine) should be a trigger to investigate for an underlying other condition. This represents a threshold above which the eGFR change is greater than would be expected from natural variation. Lastly, it should be noted that restricting salt intake may help ensure maximal effects of RASi.

Practice Point 3.6.7: Continue ACEi or ARB in people with CKD even when the eGFR falls below 30 ml/min per 1.73 m^2.

In a recent STOP-ACEi trial of 411 participants with mean eGFR of 13 ml/min per 1.73 m^2, a policy of discontinuing RASi in CKD G4–G5 did not result in any kidney or cardiovascular benefits.[507] Two observational studies have also found that associations suggesting outcomes were worse among participants who stopped RASi after reaching an

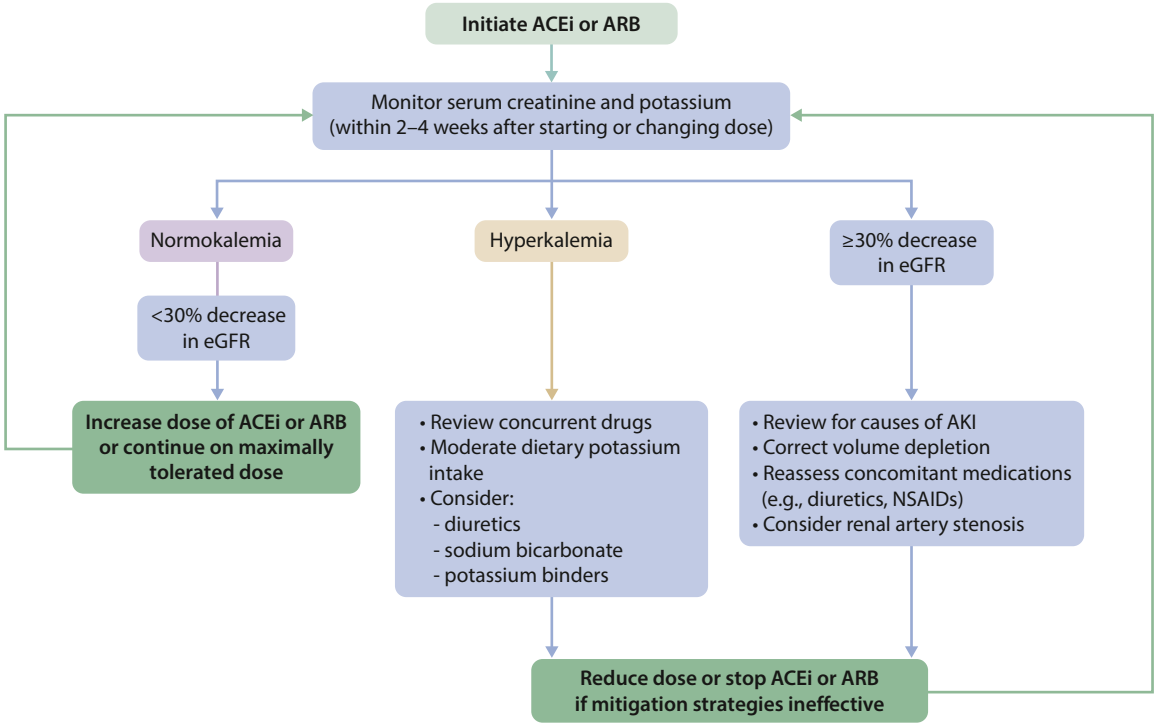

Figure 21 | Algorithm for monitoring of potassium and estimated glomerular filtration rate (eGFR) after the initiation of renin-angiotensin system inhibitors. ACEi, angiotensin-converting enzyme inhibitor; AKI, acute kidney injury; ARB, angiotensin II receptor blocker; NSAID, nonsteroidal anti-inflammatory drug. Modified from Kidney Disease: Improving Global Outcomes Diabetes Work Group. KDIGO 2022 Clinical Practice Guideline for Diabetes Management in Chronic Kidney Disease. *Kidney Int*. 2022;102:S1–S127.[23] Copyright © 2022, KDIGO: Kidney Disease Improving Global Outcomes. Published by Elsevier Inc. on behalf of the International Society of Nephrology. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

eGFR <30 ml/min per 1.73 m^2, compared with those who continue.[508,509] In addition, a recent individual patient level data meta-analysis demonstrated a benefit in delaying KRT in patients with eGFR <30 ml/min per 1.73 m^2.[510]

3.7 Sodium-glucose cotransporter-2 inhibitors (SGLT2i)

The Work Group concurs with the KDIGO 2022 Clinical Practice Guideline for Diabetes Management in Chronic Kidney Disease, which stated: "We recommend treating patients with type 2 diabetes (T2D), CKD, and an eGFR ≥20 ml/min per 1.73 m^2 with an SGLT2i *(1A)*."[23] However, in this guideline, we offer a more general 1A recommendation for adults with CKD. We also highlight practice points from the KDIGO Diabetes guideline for diabetes management in CKD, which are also relevant for people with CKD without diabetes:

Recommendation 3.7.1: We recommend treating patients with type 2 diabetes (T2D), CKD, and an eGFR ≥20 ml/min per 1.73 m^2 with an SGLT2i *(1A)*.

Practice Point 3.7.1: Once an SGLT2i is initiated, it is reasonable to continue an SGLT2i even if the eGFR falls below 20 ml/min per 1.73 m^2, unless it is not tolerated or KRT is initiated.

Practice Point 3.7.2: It is reasonable to withhold SGLT2i during times of prolonged fasting, surgery, or critical medical illness (when people may be at greater risk for ketosis).

Recommendation 3.7.2: We recommend treating adults with CKD with an SGLT2i for the following *(1A)*:
- **eGFR ≥20 ml/min per 1.73 m^2 with urine ACR ≥200 mg/g (≥20 mg/mmol), or**
- **heart failure, irrespective of level of albuminuria.**

Practice Point 3.7.3: SGLT2i initiation or use does not necessitate alteration of frequency of CKD monitoring and the reversible decrease in eGFR on initiation is generally not an indication to discontinue therapy.

Use of SGLT2i in people with T2D is recommended in previous guidelines irrespective of level of albuminuria. This new recommendation (3.7.2) places high value on the importance of reducing risk of kidney failure, cardiovascular mortality, and heart failure in people with CKD and high value on the large relative reductions in risk for kidney disease progression in a series of large, placebo-controlled RCTs. It also places moderate value on the benefits of SGLT2i on risk of AKI, hospitalization for heart failure and myocardial infarction, risk of hospitalization from any cause, and high value on the demonstrable net absolute benefits versus absolute harms in people with CKD (particularly in those

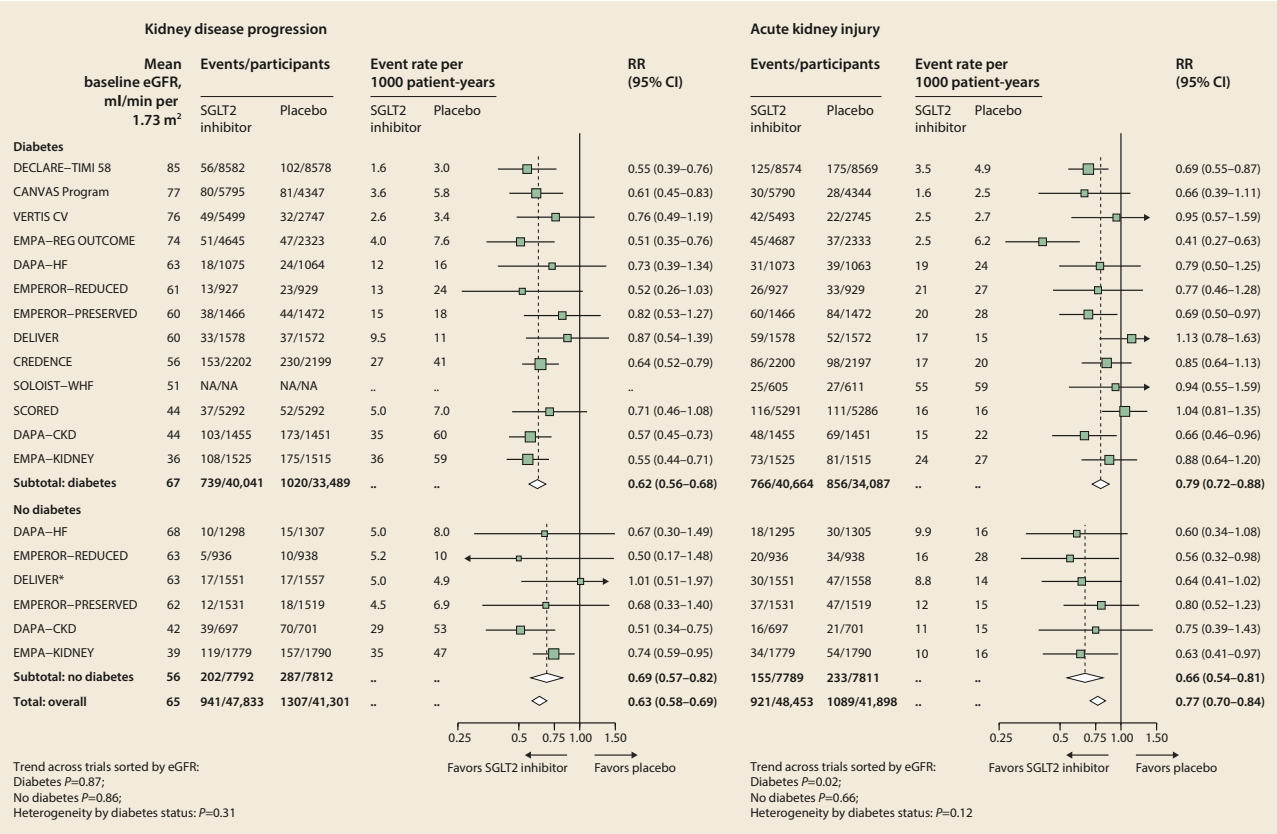

Figure 22 | Effect of sodium-glucose cotransporter-2 inhibitors (SGLT2i) with kidney disease outcomes by diabetes status. ACR, albumin-to-creatinine ratio; CI, confidence interval; eGFR, estimated glomerular filtration rate; NA, not applicable; RR, relative risk. Reproduced from Nuffield Department of Population Health Renal Studies Group, SGLT Inhibitor Meta-Analysis Cardio-Renal Trialists' Consortium. Impact of diabetes on the effects of sodium glucose co-transporter-2 inhibitors on kidney outcomes: collaborative meta-analysis of large placebo-controlled trials. *Lancet*. 2022;400:1788–1801.[511] © 2022 The Author(s). Published by Elsevier Ltd. This is an Open Access article under the CC BY 4.0 license.

without diabetes who are at very low risk of ketoacidosis). SGLT2i also favorably reduce BP, uric acid levels, measures of fluid overload, the risk of serious hyperkalemia, and do not increase risk of hypoglycemia. The recommendation is consistent with but expands on Recommendation 1.3.1 from the KDIGO 2022 Clinical Practice Guideline for Diabetes Management in Chronic Kidney Disease *to include people with causes of CKD not related to diabetes*.

Key information

Balance of benefits and harms. *Benefits.* Several large, placebo-controlled RCTs have provided clear demonstrations of the efficacy of SGLT2i, which substantially reduce the risk of kidney failure, AKI, and hospitalization for heart failure, and also moderately reduce the risk of cardiovascular death and myocardial infarction in people with and without CKD. These benefits appear to be irrespective of diabetes status, cause of kidney disease, or level of GFR.[511,512] The benefits of SGLT2i in people with diabetes and CKD have been fully described in the KDIGO 2022 Clinical Practice Guideline for Diabetes Management in Chronic Kidney Disease.[23]

Two large RCTs using 2 different SGLT2i recruited 10,913 participants and focused on CKD populations at risk of progression, reporting benefits in terms of kidney disease progression.[403,513] Key differences between the 2 trials were the inclusion of a large number of causes of kidney disease not related to diabetes, lower eGFR, and lower levels of ACR in The Study of Heart and Kidney Protection With Empagliflozin (EMPA-KIDNEY) compared with the Dapagliflozin and Prevention of Adverse Outcomes in Chronic Kidney Disease (DAPA-CKD) trial.

In a collaborative metanalysis including those 2 and 11 other trials (13 trials with just over 90,000 randomized participants) in comparison with placebo, those allocated to an SGLT2i experienced a 37% reduction in the risk of kidney disease progression and a 23% reduction in the risk of AKI irrespective of diabetes status (Figure 22).[511]

The same meta-analysis showed that, compared with placebo, allocation to an SGLT2i reduced the risk of the composite of cardiovascular death or hospitalization for heart failure by 23% irrespective of diabetes status (Figure 23[511]), although there were limited numbers of cardiovascular events in people with CKD without diabetes. SGLT2i also afford an approximate 10% RR reduction in MACE, primarily from reduced risk of cardiovascular death and myocardial infarction with no clear effect on stroke.[511,512]

Furthermore, SGLT2i also importantly reduce the risk of hospitalization from any cause,[403] reduce BP,[403,513,514] uric

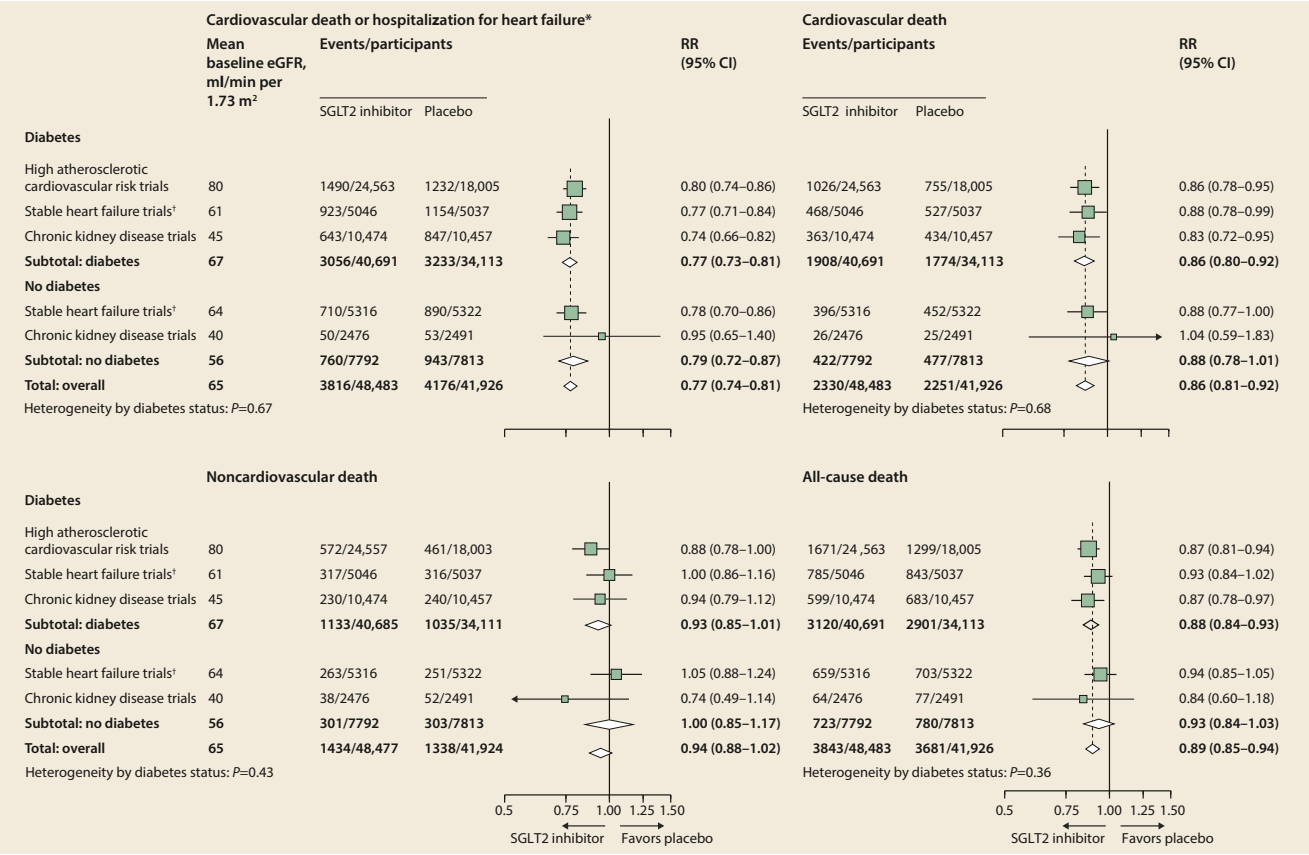

Figure 23 | Effects of sodium-glucose cotransporter-2 (SGLT2) inhibition versus placebo on cardiovascular and mortality outcomes by diabetes status and trial population. Collaborative meta-analysis of data from 13 large placebo control trials of SGLT2 inhibitors. †Data from Effect of Sotagliflozin on Cardiovascular Events in Patients with Type 2 Diabetes Post Worsening Heart Failure (SOLOIST-WHF) are included in totals but excluded from the stable heart failure trials group as the trial included patients with recent acute decompensated heart failure. CI, confidence interval; eGFR, estimated glomerular filtration rate; RR, relative risk. Reproduced from Nuffield Department of Population Health Renal Studies Group, SGLT Inhibitor Meta-Analysis Cardio-Renal Trialists' Consortium. Impact of diabetes on the effects of sodium glucose co-transporter-2 inhibitors on kidney outcomes: collaborative meta-analysis of large placebo-controlled trials. *Lancet*. 2022;400:1788–1801.[511] © 2022 The Author(s). Published by Elsevier Ltd. This is an Open Access article under the CC BY 4.0 license.

acid levels,[515] weight/fluid overload,[516] and the risk of serious hyperkalemia.[517]

Harms. SGLT2i are well tolerated with high levels of adherence in the RCTs in CKD.[403,513,514] In the studied populations, any risk of ketoacidosis or lower-limb amputation resulting from SGLT2i use was substantially lower than the potential absolute benefits and generally restricted to people with diabetes. Meta-analysis estimates of absolute benefits and harms for each 1000 people with CKD and T2D treated for 1 year with an SGLT2i were 11 fewer cardiovascular deaths or hospitalizations for heart failure, for approximately 1 episode of ketoacidosis and approximately 1 lower-limb amputation, respectively (and also 11 fewer people developing kidney disease progression and 4 fewer people with AKI). The corresponding benefits in people with CKD without diabetes were 15 fewer people with kidney disease progression, 5 fewer with AKI, and 2 fewer cardiovascular deaths or hospitalizations for heart failure per 1000 patient-years of treatment with no excess risk of ketoacidosis or amputation observed.[511] The vast majority of urinary tract infections in people taking SGLT2i are not caused by SGLT2 inhibition, and there is no increased risk of hypoglycemia. There is an increased risk of mycotic genital infections (in men and women), but these are generally mild and treating these infections with low-cost topical agents should help treatment adherence.

Certainty of evidence. SGLT2i have been studied in a series of large trials with consistent effects observed between trials, using different agents in the class. The trials have robust double-blind designs that minimize risk of bias, and they have provided precise estimates of effect with no risk of publication bias due to the Nuffield Department of Public Health (NDPH) Renal Studies Group and SGLT2 inhibitor Meta-Analysis Cardio-Renal Trialists' Consortium (SMART) collaboration, which brought together all the trialists that have conducted the relevant large trials. The totality of the evidence provides high levels of certainty of efficacy, with larger effect sizes observed in many populations. Relative effects on kidney disease progression appeared to be larger among people with higher levels of albuminuria who are at highest absolute risk of progression. The size of RR reductions

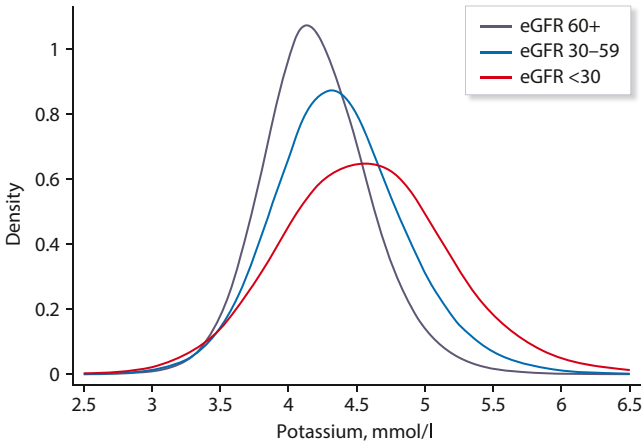

Figure 29 | Distribution of blood potassium in general population and high-risk cohorts from the Chronic Kidney Disease Prognosis Consortium, by estimated glomerular filtration rate (eGFR). Density refers to the proportion of the population experiencing serum potassium level (e.g., 0.08 of the population with a GFR >60 have a potassium of 3.8; conversely, 0.2 of the population with a GFR <30 have a potassium of 5.5). Reproduced from Kovesdy CP, Matsushita K, Sang Y, et al. Serum potassium and adverse outcomes across the range of kidney function: a CKD Prognosis Consortium meta-analysis. *European Heart Journal* 2018;39:1535–1542 by permission of Oxford University Press on behalf of the European Society of Cardiology.[555] All rights reserved. © The Author(s) 2018. Inclusion under a Creative Commons license is prohibited. https://doi.org/10.1093/eurheartj/ehy100

levels of GFR, thus understanding potassium physiology and its impacting factors are important in effective patient care.

Hyperkalemia in people with preserved GFR is less prevalent. An acute episode of hyperkalemia is a potassium result above the upper limit of normal that is not known to be chronic. At the current time, there is no consensus on the magnitude, duration, and frequency of elevated potassium values that define chronicity.[556] In addition to decreased eGFR, other risk factors for hyperkalemia included higher ACR and prior diabetes, hyperglycemia, constipation, RASi,[557] and MRA.[536] Note that SGLT2i do not appear to increase serum potassium values.[403,517]

Studies have demonstrated a continuous U-shaped relationship between serum potassium and all-cause mortality in a range of different populations (Figure 31).[555,558] It has also been associated with worse kidney prognosis.[559] Observationally, the risk of death from the same degree of hyperkalemia is lower in more advanced CKD stages.[560–564] This may suggest that there are adaptive mechanisms that render better tolerance to elevated levels of potassium in circulation.[561,565–568]

3.11.1 Awareness of factors impacting on potassium measurement

There are several factors and mechanisms that may impact on potassium measurements, including the actions of medications that can increase the risk of developing hyperkalemia. These are summarized in Tables 25[556,569–575] and 26.[23,576]

Practice Point 3.11.1.1: Be aware of the variability of potassium laboratory measurements as well as factors and mechanisms that may influence potassium measurement including diurnal and seasonal variation, plasma versus serum samples, and the actions of medications.

The Work Group would like to highlight Figure 26 for the monitoring of serum potassium during treatment with a nonsteroidal MRA (finerenone) from the KDIGO 2022 Clinical Practice Guideline for Diabetes Management in Chronic Kidney Disease.[23]

Hyperkalemia has been associated with therapeutic actions of either reducing or stopping RASi.[577–580] Steps can be taken to mitigate the risk of hyperkalemia and improve potassium control that could increase the use of RASi in people with an evidenced indication. For details on how to manage hyperkalemia associated with the use of RASi and associated

No diabetes

	Hyperkalemia		
eGFR	A1	A2	A3
>90	1.5% (0.4, 4.6)	1.1% (0.3, 3.2)	1.4% (0.4, 4.4)
75–89	1.7% (0.5, 5.1)	1.6% (0.5, 4.8)	1.5% (0.5, 4.7)
60–74	2.3% (0.7, 7.0)	2.0% (0.6, 6.0)	2.3% (0.7, 7.0)
45–59	**4.5% (1.4, 12.8)**	3.5% (1.1, 10.3)	5.2% (1.6, 14.6)
30–44	9.5% (3.0, 24.8)	10.5% (3.3, 26.9)	11.3% (3.6, 28.5)
15–29	16.1% (5.3, 37.5)	19.0% (6.4, 42.5)	23.7% (8.3, 49.4)

Diabetes

	Hyperkalemia		
eGFR	A1	A2	A3
>90	1.8% (0.5, 5.5)	3.6% (1.1, 10.5)	1.0% (0.3, 3.0)
75–89	2.8% (0.8, 8.3)	4.0% (1.2, 11.6)	4.6% (1.4, 13.0)
60–74	3.9% (1.2, 11.2)	5.0% (1.5, 14.2)	6.6% (1.8, 17.4)
45–59	**8.8% (2.7, 23.3)**	9.9% (3.1, 25.5)	11.4% (3.6, 28.8)
30–44	12.8% (4.1, 31.5)	18.7% (6.3, 41.9)	87.5% (67.1, 95.6)
15–29	24.7% (8.8, 50.8)	31.5% (11.9, 59.1)	34.4% (13.3, 62.2)

Figure 30 | Meta-analyzed adjusted prevalence of hyperkalemia (25th and 75th percentile cohort) in general population and high-risk cohorts from the Chronic Kidney Disease Prognosis Consortium, by diabetes status. Hyperkalemia is defined as potassium >5 mmol/l. The adjusted prevalence of hyperkalemia at each estimated glomerular filtration rate (eGFR) and albuminuria stage was computed as follows: first, the random-effects weighted adjusted mean odds at the reference point (eGFR 50 ml/min per 1.73 m^2) was converted into a prevalence estimate. To the reference estimate, the meta-analyzed odds ratios for hyperkalemia were applied to obtain prevalence estimates at eGFR 95, 80, 65, 35, and 20 ml/min per 1.73 m^2 for each stage of albuminuria. The prevalence estimates were adjusted to 60 years old, half male, non-Black, 20% history of CVD, 40% ever smoker, and body mass index 30 kg/m^2. The 25th and 75th percentiles for predicted prevalence were the estimates from individual cohorts in the corresponding percentiles of the random-effects weighted distribution of adjusted odds. A1, albuminuria <30 mg/g (<3 mg/mmol); A2, albuminuria 30–300 mg/g (3–30 mg/mmol); A3, >300 mg/g (>30 mg/mmol). Reproduced from *American Journal of Kidney Diseases*, volume 73, issue 2, Inker LA, Grams ME, Levey AS, et al. Relationship of estimated GFR and albuminuria to concurrent laboratory abnormalities: an individual participant data meta-analysis in a Global Consortium, pages 206–217, Copyright © 2018, with permission from the National Kidney Foundation, Inc.[541]

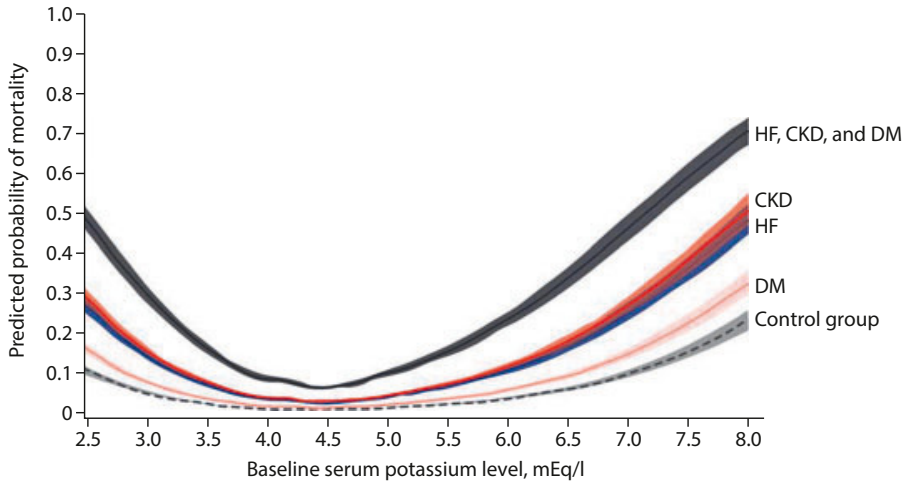

Figure 31 | Serum potassium concentration and confounder-adjusted risk of death by the presence or absence of diabetes, heart failure (HF), or chronic kidney disease (CKD). Reproduced from Collins AJ, Pitt B, Reaven N, et al. Association of serum potassium with all-cause mortality in patients with and without heart failure, chronic kidney disease, and/or diabetes. *Am J Nephrol*. 2017;46:213–221.[558] © 2017 The Author(s) Published by S. Karger AG, Basel. This article is licensed under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License (CC BY-NC-ND) (http://www.karger.com/Services/OpenAccessLicense).

Table 25 | Factors and mechanisms that impact on potassium measurements[556,569-575]

Factor/mechanism	Possible cause/clinical implication

Pseudohyperkalemia—*in vivo* serum potassium is normal and commonly GFR preserved, but during the process of drawing blood or clotting, there has been a release of intracellular potassium	• Tight tourniquet
	• Hand/arm exercising or clenching at the time of blood draw
	• Hemolysis due to vigorous shaking of blood vial/inappropriate blood draw equipment/inappropriate storage of samples
	  • If suspected, blood should be retaken and analyzed in the appropriate manner and time frame[556,569]
	• Presence of thrombocytosis/leukocytosis
	  • If suspected, take plasma potassium as serum potassium may be falsely increased[570]

Hyperkalemia due to disruption in the mechanism of shifting potassium out of cells	• Increase in plasma osmolarity (e.g., dehydration and hyperglycemia) • Massive tissue breakdown (e.g., rhabdomyolysis and tumor lysis syndrome) • β-adrenergic blockade, especially during and immediately after exercise[569] • Insulin deficiency • Aldosterone blockade • Nonorganic acidosis
Hyperkalemia due to disruption in the mechanism of moving potassium into cells	• Disruption in the release of insulin in response to raised serum potassium (e.g., in uncontrolled diabetes) • Disruption to the release of aldosterone in response to a raised serum potassium[569]
Hyperkalemia due to the decreased ability to excrete potassium	• Advancing CKD resulting in inability to excrete excessive potassium • Constipation: in advancing CKD, the gut assumes a much more important role in maintaining potassium balance by increasing the excretion of potassium[571,572] • Medications: blocking the RAAS pathway and other medication resulting in the inability to excrete excessive potassium (Table 26)[569,573]
Diurnal variation in potassium excretion with most excretion in humans occurring close to noon	• Circadian excretion of kidney electrolytes have been well documented.[574] Clinical relevance is yet to be understood • Note the 0.24–0.73 mmol/l variation in K+ values within individuals over a 24-hour period
Plasma vs. serum potassium values	• Potassium values differ between serum and plasma values with serum values being typically higher. Healthcare providers need to be aware of the right reference values for the sample[570]
Postprandial hyperkalemia	• As kidney function declines in CKD, there is a corresponding decline in the ability of the kidneys to increase kaliuresis postprandially, eventually becoming insufficient to maintain external potassium balance[575]

CKD, chronic kidney disease; GFR, glomerular filtration rate; K+, potassium; RAAS, renin-angiotensin-aldosterone system.

Table 26 | Medications associated with increased risk of hyperkalemia

Class	Mechanism	Example
ACEi	Inhibit conversion of angiotensin I to angiotensin II	Captopril, lisinopril, perindopril, etc.
ARB	Inhibit activation of angiotensin I receptor by angiotensin II	Losartan, irbesartan, candesartan, etc.
Aldosterone antagonist	Block aldosterone receptor activation	Spironolactone, eplerenone, and finerenone
β-Adrenergic receptor blocker	Inhibit renin release	Propranolol, metoprolol, and atenolol
Digitalis glycoside	Inhibit Na^+-K^+-ATPase, necessary for collecting duct K^+ secretion	Digoxin
Heparin	Reduced production of aldosterone	Heparin sodium
Potassium-sparing diuretic	Block collecting duct apical Na^+ channel, decreasing gradient for K^+ secretion	Amiloride and triamterene
NSAIDs	Inhibit synthesis of prostaglandin E and prostacyclin, inhibiting renin release	Ibuprofen, naproxen, diclofenac, etc.
CNI	Inhibit Na^+-K^+-ATPase, necessary for collecting duct K^+ secretion	Cyclosporine and tacrolimus
ns-MRA	Block MR-mediated Na^+ reabsorption	Finerenone
Other	Block collecting duct apical Na^+ channel, decreasing gradient for K^+ secretion	Trimethoprim and pentamidine

ACEi, angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; ATP, adenosine triphosphate; CNI, calcineurin inhibitor; K^+, potassium; Na^+, sodium; NSAID, nonsteroidal anti-inflammatory drug; ns-MRA, nonsteroidal mineralocorticoid receptor antagonist.
Data from Weiner *et al.*[576] and Kidney Disease: Improving Global Outcomes Diabetes Work Group.[23]

monitoring, please refer to Figure 21. See Section 4.3 for more information on continuing RASi after hyperkalemia events.

3.11.2 Potassium exchange agents

Practice Point 3.11.2.1: Be aware of local availability or formulary restrictions with regard to the pharmacologic management of nonemergent hyperkalemia.

The pharmacologic management of nonemergent hyperkalemia has a number of clinical tools with the increased number of licensed potassium exchange agents. These medications have differing mechanisms of action, onsets of clinical effects, and potential medication and disease-state interactions (Table 27[581,582]). Although the classic potassium exchange agents have had tolerability issues, the newer ones appear to have fewer such issues and appear relatively safe when used long term.[575,583,584] Use of these newer exchange agents may help facilitate essential use of RASi/MRA. However, it is important that the healthcare provider be aware of clinical nuances and local availability or formulary restrictions in determining therapy selection.[583] A comparison of available potassium exchange agents can be found in Table 27.

3.11.3 Timing to recheck potassium after identifying moderate and severe hyperkalemia in adults

"Think Kidneys" and the UK Kidney Association have provided a practical guide, which we have adapted (Table 28) for repeat testing after a hyperkalemic episode.[585] The timing of repeat testing is guided by the level of hyperkalemia and the clinical context.[586]

3.11.4 Managing hyperkalemia

In people with CKD and the management of nonemergent hyperkalemia, a systematic approach of treating correctable factors (e.g., correction of severe metabolic acidosis) and understanding the role of diet and medications may provide a pragmatic framework. Figure 32 shows a stepwise practical approach to the management of hyperkalemia in CKD.

3.11.5 Dietary considerations

In early stages of CKD, high intake of foods naturally rich in potassium appears to be protective against disease progression, and dietary restriction of foods naturally containing potassium, such as fruits and vegetables, may be harmful to cardiac health; therefore, such restriction is not endorsed.[587]

Practice Point 3.11.5.1: Implement an individualized approach in people with CKD G3–G5 and emergent hyperkalemia that includes dietary and pharmacologic interventions and takes into consideration associated comorbidities and quality of life (QoL). Assessment and education through a renal dietitian or an accredited nutrition provider are advised.

Practice Point 3.11.5.2: Provide advice to limit the intake of foods rich in bioavailable potassium (e.g., processed foods) for people with CKD G3–G5 who have a history of hyperkalemia or as a prevention strategy during disease periods in which hyperkalemia risk may be a concern.

Diet may increase serum potassium postprandially,[575,588] but other conditions such as the use of potassium-sparing

Table 27 | A comparison of potassium exchange agents

	(Polystyrene sulfonates) sodium or calcium	Patiromer	Sodium zirconium cyclosilicate (SZC)
Mechanism of action	Sodium-potassium exchange resin (SPS) or calcium-potassium exchange resin (CPS)	Calcium-potassium exchange polymer	Crystalline compound that traps K^+ in exchange for hydrogen and sodium cations
Counterion content	SPS: Suspension contains 65 mmol/60 ml (15 g) of sodium and powder approximately 4.1 mmol/g of sodium. CPS: 1.6–2.4 mmol/g of calcium.	1600 mg of calcium per 8.4 g of patiromer	Approximately 400 mg of sodium per 5 g of SZC
Cations bound	Potassium, magnesium, and calcium	Potassium, magnesium, and phosphate (bound by calcium release)[582]	Potassium
Formulation of route of administration	Powder for reconstitution (oral), suspension (oral), and enema (rectal)	Powder for reconstitution (oral)	Powder for reconstitution (oral suspension)
Dosage and titration	Oral: 15–60 g/d (up to 4 times per day) Rectal: 30 g/d (for SPS up to a maximum of 50 g/d)	Initial: 8.4 g orally once per day (maximum 25.2 g orally once per day); dose can be increased by 8.4 g increments at 1-week intervals	Initial: 10 g orally 3 times per day for up to 48 hours
Maintenance dosing	15–60 g/d orally per day depending on potassium level and level of tolerability	8.4–25.2 g orally once per day	5 g every second day to 10 g once per day
Onset of effect	Variable, hours to days	4–7 hours	Starts to reduce potassium within 1 hour with normokalemia typically at 24–48 hours
Duration of effect	Variable, 6–24 hours	24 hours	Not studied; not systematically absorbed and excreted fecally
Administration pearls	Separate from oral medications by at least 3 hours before or 3 hours after administration; if gastroparesis, separate other medications by 6 hours	Separate from oral medications by at least 3 hours before or 3 hours after administration except for those drugs to not have a clinically important interaction	No dose adjustments or separation of time of dosing is required for any medication that does not have pH-dependent bioavailability. However, SZC should be administered at least 2 hours before or 2 hours after oral medicinal products with clinically meaningful gastric pH-dependent bioavailability
Adverse effects	GI events (nausea, vomiting, diarrhea, constipation), electrolyte disturbances (hypokalemia, hypocalcemia, and hypomagnesemia), edema, and potentially serious GI adverse events (intestinal necrosis, bleeding, ischemic colitis, and perforation)	GI events (nausea, diarrhea, and flatulence), electrolyte disturbances (hypokalemia, hypercalcemia, and hypomagnesemia) Insufficient postmarketing surveillance at present to evaluate long-term/rare events	Hypokalemia and edema events are the most common. Milder reports of GI events (nausea, diarrhea, and constipation) Insufficient postmarketing surveillance at present to evaluate long-term/rare events

GI, gastrointestinal.
Modified from Bridgeman MB, Shah M, Foote E. Potassium-lowering agents for the treatment of nonemergent hyperkalemia: pharmacology, dosing and comparative efficacy. *Nephrology Dialysis Transplantation*, Volume 34, Supplement 3, pages iii45–iii50.[581] © The Author(s) 2019. Published by Oxford University Press on behalf of ERA-EDTA. All rights reserved.

Table 28 | Suggested action in the event of moderate and severe hyperkalemia

Severity of hyperkalemia	Clinically unwell or AKI	Unexpected result
Moderate K^+ 6.0–6.4 mmol/l	Assess and treat in hospital	Repeat within 24 hours
Severe K^+ ≥6.5 mmol/l	Take immediate action to assess and treat. Assessment will include blood testing and electrocardiogram monitoring	

AKI, acute kidney injury; K^+, potassium.
Data from Think Kidneys, the Renal Association and the British Society for Heart Failure.[585]

medications, metabolic acidosis, hyperosmosis due to hyperglycemia, hypernatremia or uremia, and constipation are more likely to explain plasma potassium abnormalities than diet.[545,556,572,589] Although short-term dietary restriction of the foods highest in potassium is a valid strategy to treat acute hyperkalemia, restriction of foods highest in bioavailable potassium may be a supportive prevention strategy for people with a history of hyperkalemia or during periods in which hyperkalemia risk is a concern.[590] Increased efforts toward education on potassium content in foods can serve to improve diet quality and diversity for many people with CKD where this

1st line: **Address correctable factors**	• Review non-RASi medications (e.g. NSAIDs, trimethoprim) • Assess dietary potassium intake (dietary referral) and consider appropriate moderation of dietary potassium intake
2nd line: **Medications**	Consider: • Appropriate use of diuretics • Optimize serum bicarbonate levels • Licensed potassium exchange agents
3rd line: **Last resort**	• Reduce dose or discontinue RASi/MRA (Discontinuation is associated with increased cardiovascular events. Review and restart RASi or MRA at a later date if patient condition allows.)

Figure 32 | Actions to manage hyperkalemia (potassium >5.5 mmol/l) in chronic kidney disease. MRA, mineralocorticoid receptor antagonists; NSAID, nonsteroidal anti-inflammatory drug; RASi, renin-angiotensin system inhibitors.

restriction may not be needed.[546,556,591] Although guidelines and available information to people with CKD have heavily emphasized plant-based foods as potential causes of hyperkalemia in CKD,[592] other healthy nutrients in plant-based foods affect potassium absorption and distribution[588,593,594]; therefore, the net bioavailable potassium from plant-based foods may be lower than appreciated.[595] Highly processed foods (rich in potassium additives), meats, dairy products, juices, and salt substitutes made with potassium chloride are actually higher in absorbable potassium than many plant-based fresh foods (Figure 33[592]).[596–598]

Teaching materials for people with CKD should place a greater focus on highly processed versus unprocessed food restriction for hyperkalemia management.[592] An example of a patient resource for potassium management can be found at: http://www.bcrenal.ca/resourcegallery/Documents/Potassium_Management_in_Kidney_Disease.pdf.

Cooking methods such as soaking foods for 5–10 minutes in previously boiled water can effectively reduce the potassium by half for some foods.[599] Thus, educating people with CKD and healthcare providers, using clear messaging, on dietary approaches to potassium management is needed (https://www.theisn.org/initiatives/toolkits/raasi-toolkit/#1684867542809-330edb79-52b4), as well as a policy to improve food labeling by detailing the added potassium used in processing.

Special considerations

International considerations. For people with CKD and severe recurrent hyperkalemia (potassium >6 mmol/l), the balance to be considered is between the additional cost of the number needed to treat with potassium exchange agents to prevent additional costs of hyperkalemia over and above CKD management costs. If the price for potassium-lowering therapy is lower than the reduction of inpatient and outpatient costs due to prevented hyperkalemia, the cost-benefit ratio will be favorable because in addition to the health benefits, there is a net saving of healthcare costs resulting from potassium-lowering treatment. The key is to implement a successful affordable strategy for hyperkalemia management that allows the maintenance of other therapies

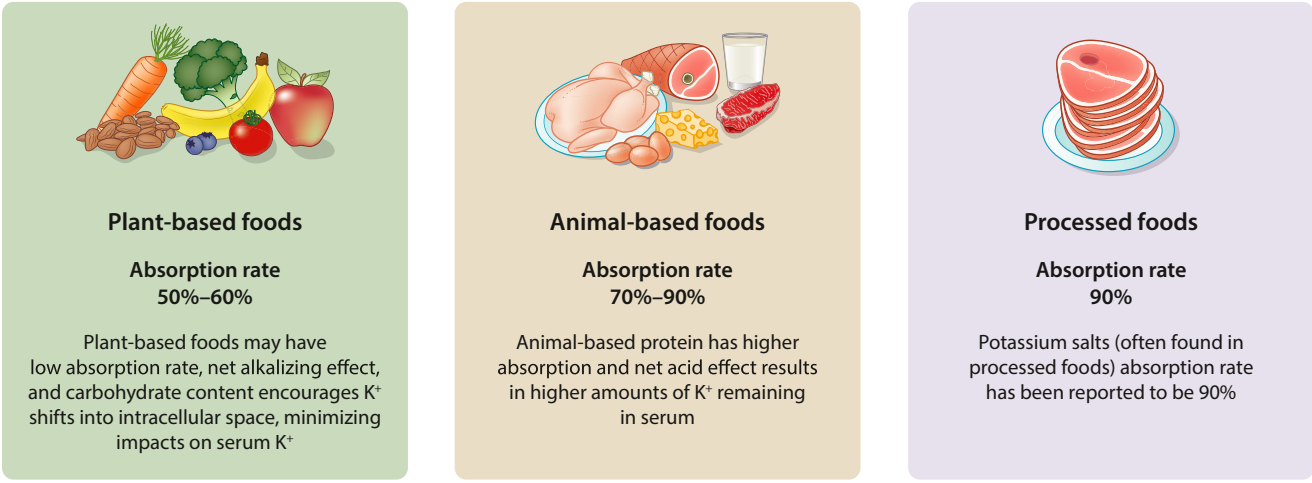

Figure 33 | Potassium absorption rates of plant-based, animal-based, and processed foods. Data from Picard K, Griffiths M, Mager DR, Richard C. Handouts for low-potassium diets disproportionately restrict fruits and vegetables. *J Ren Nutr*. 2021;31:210–214.[592]